DES Pune University, Pune

Academic Year: 2026-27	**School of Engineering and Technology**
Semester: Even	**Electronics and communication Engineering (AIML)**
Class & Div: TY-A & B	

PBEL Logbook Record

S.No.	Date (From-To)	Planned Activities	Work Carried Out	Tools / Techniques Used	Observations / Results	Issues / Challenges	Action Plan for Next Week	Student Signature	Mentor Remarks & Signature
1	Week 1	Project definition, feasibility study, and literature review.	Finalized project scope: ECG Arrhythmia classification using SNNs on FPGA. Conducted literature review of 20+ IEEE papers to benchmark metrics.	IEEE Xplore, Mendeley, Web searching for benchmark datasets.	Identified MIT-BIH database as gold standard. Selected multiplier-free LIF and Rate Coding as target hardware.	Finding papers targeting FPGA without heavy DSP slices was difficult.	Download MIT-BIH dataset and begin Python data preprocessing.		
2	Week 2	Data acquisition, formatting, and initial exploration.	Downloaded MIT-BIH database. Wrote preprocess.py to extract 128-sample windows around R-peaks. Mapped to 5 AAMI classes. Generated exploration heatmaps.	Python, wfdb library, NumPy, Matplotlib.	Successfully extracted ~100k heartbeats. Normalization to [0, 1] range successfully preserves waveform shape for hardware.	Noticed severe class imbalance (Normal beats heavily outnumber Arrhythmia beats).	Implement data augmentation and class weighting to fix dataset imbalance.		
3	Week 3	SNN Python Architecture & Data Balancing.	Built augment_data.py for minority classes. Developed train_snn.py using 128->16->16->5 architecture with Surrogate Gradient Learning.	Python, PyTorch, snnTorch, BPTT.	Training script handles temporal dynamics over 16 timesteps. Augmented data improves minority class visibility.	Initial training yielded 26.8% accuracy due to thresholding and imbalance issues.	Tune hyperparameters (threshold, LR) and begin translating SNN logic to VHDL.		
4	Week	Hardware Design	Designed and coded core	VHDL, Fixed-	Successfully replaced	Ensuring VHDL	Complete weight		

	4	(VHDL) Phase 1: Core Modules.	hardware modules: snn_pkg.vhd (types), lfsr_8bit.vhd (Rate Encoder), argmax_5.vhd (Combinational classifier), and snn_top.vhd (FSM).	point arithmetic (Q8.8), FSM methodology.	hardware multipliers with bit-shift leak mechanisms. FSM successfully time-multiplexes neurons.	reset mechanism matches Python model's "zero reset" exactly required careful state logic.	quantization script and write VHDL Testbench.		
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